3D print job: Alvik robot capstone (Nick)

Print 2 of every file. 7 STL files, 14 parts. All PLA, 0.4 mm nozzle, 0.2 mm layers, **no supports**.

Settings that matter for size

1. **Units are millimetres. Import at 100% scale.** Check the slicer shows the sizes in the table below. If a part shows about 25 times too big or too small, it was imported in inches: re-import as mm.

2. **Don't rotate the parts.** Each file already lies on the bed the way it needs, with no supports.

3. **Leave XY size compensation (horizontal expansion) at 0**, the default. The nose bracket screws into the robot and the dock's pocket fits that bracket with only 0.5 mm of play, so the sizes must come out as drawn.

4. **The dock is 196 mm long.** With its brim it needs a bed at least about 210 mm in one direction (fine on a Prusa MK3/MK4, Bambu X1/P1/A1, Ender 3; it will NOT fit an A1 mini or other 180 mm beds).

Parts

File	Qty	Size as it lies X x Y x Z mm	Colour	Settings	PLA each (about)
dock.stl	2	110 x 196 x 22	Black	4 walls, 20% infill, brim on	60 g
nose_bracket.stl	2	38 x 102 x 31	Any	4 walls, 20% infill (it is screwed on)	15 g
check_tile.stl	2	40 x 40 x 1.8	White, then black	Filament change (pause / M600) at 1.2 mm height so the raised squares print black. No colour change? Print white; the squares get coloured with a marker.	3 g
extruder.stl	2	162 x 40 x 75	Any except white	2-3 walls, 5-10% infill (lightning if available)	70-100 g
strander.stl	2	150 x 56 x 56	Any except white	2-3 walls, 5-10% infill	75-110 g
takeup_reel.stl	2	100 x 64 x 68	Any except white	2-3 walls, 5-10% infill	55-85 g
test_bay.stl	2	140 x 140 x 60	Any except white	2-3 walls, 5-10% infill	100-160 g

Total: about 1 kg of PLA for all 14 parts; the four machine models are most of it. Print the dock, nose brackets and check tiles first (about 150 g in all); the machines can follow.

Why "any colour except white": the overhead camera finds the robots partly by their white tops, so the machine models must not be white. A dark or strong colour (grey, blue, red) is best.

For Nick (not the print tech)

Parts to buy for the nose brackets, per robot: 2 Pololu #1528 mini lever switches, 4 M3 x 8 mm button-head screws, 4 M2 x 8 mm self-tapping screws, 1 Grove-to-female-jumper cable, 4 short jumper wires.

Not in this batch: the LED holder (being redesigned next week).